



18 April 2025

International Journal of Geographical Information Science

Dear Editor-in-Chief Professor May Yuan:

We are pleased to submit our manuscript entitled "Measuring causal associations by geographical cross mapping cardinality".

This study proposed a novel Geographical Cross Mapping Cardinality (GCMC) model developed to measure causal associations from spatial cross-sectional data, addressing a critical gap in Earth system science, where there is a lack of data-driven models capable of examining causal associations from a spatial perspective.

The developed GCMC model provides a reliable methodological framework for measuring causal associations and identifying the causal link, with broad applicability across various fields. We selected three case studies to demonstrate the application potential of GCMC in the manuscript, while also testing its sensitivity to noise. We believe that our proposed new model can provide a significant contribution to the field of GIS and spatial analysis. We look forward to the opportunity to contribute to this journal.

Author contributions: Wenbo Lv, Shaoqing Dai, and Yongze Song conceived the idea and designed the study; Wenbo Lv and Wufan Zhao collaborated on the implementation of the GCMC model in this manuscript; Wen Yi, Yumiao Xiao, and Nan Jia collected and analyzed the data; Wenbo Lv and Yongze Song wrote the first draft, and all authors contributed to the writing and editing of the final manuscript.

The authors declare no competing interests.

Please do not hesitate to contact me should you require additional information. We appreciate your kind consideration.

Sincerely yours,
Yongze

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